ADITYA SINGH

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EDUCATION

Parul University **April 2025**
Bachelors of Technology, Computer Science- Specialization in AI & ML *Vadodara,Guj.*

- Coursework: Machine Learning, Data Structures and Algorithms, Artificial Intelligence, Deep Learning, NLP

GATE DS & AI Qualified **May 2025**

EXPERIENCE

Associate Engineer (AI) **May 2025 – Present**
Capsitech *Jodhpur, Raj.*

- Engineered **an NLP-driven assistant** for internal documentation search, cutting lookup **time by 40%.**
- Led integration **of AI microservices** into existing platforms using **Docker** and **RESTful APIs**, ensuring seamless production rollout.
- Implemented modular AI microservices with Docker, FastAPI, and Kubernetes, enabling scalable, low-latency inference (<200ms).

AI Intern **Feb 2025 – May 2025**
Siimteq Technologies *Vadodara, Guj.*

- Developed a custom chatbot using **fine-tuned BERT with attention masking**, achieving 91% precision in intent classification across 50+ user intents.
- Reduced model response **time by 40%** by implementing **asynchronous inference** and **response caching with Redis** in the FastAPI backend.
- Enhanced **chatbot flow** using custom **dialog states, confidence thresholds**, and fallback strategies to improve response quality.

PROJECTS

Document Analyzer [GitHub]

- Parsed and analyzed **~120 scientific PDFs**, extracting ~12,000 sentences and performing NER using BERT and spaCy pipelines.
- Identified **900+ mathematical expressions** using rule-based pattern matching and spatial clustering with an estimated **~89% detection accuracy**.
- Combined **BERT extractive** summarization with **LLaMA 3.3** abstractive generation, achieving compression ratios of **40–55%** with strong semantic coherence.

AI-Powered Customer Support Chatbot [GitHub]

- Designed a scalable chatbot using Dialogflow CX and BERT embeddings, trained on **1,000+ user phrases across 50+ intents**, achieving **90% intent classification accuracy**.
- Refined fallback logic using **cosine similarity** rerouting and trigram matching, lowering unmatched queries **by ~18%.**
- Served via FastAPI backend with Redis caching, supporting **~400 queries/day** at **<300ms average latency**.

Emotion Detection [GitHub]

- Built a 6-class text emotion classifier using BiLSTM with TF-IDF and word2vec embeddings, trained on **100,000+ labeled text samples**, achieving **85% accuracy**.
- Boosted **F1-score by 9%** using dropout, **z-score** normalization, and grid-tuned hyperparameters.
- Served the model via Flask with multithreading, **maintaining ~0.8s average** response time under load.

SKILLS

Languages: Python, SQL, NoSQL
Frameworks: AutoGen, LangChain, Spark, PowerBI
Developer Tools: Git, Github, Gitlab, JIRA, Docker, AWS, MLFlow
Libraries: PyTorch, TensorFlow, Keras, HuggingFace, Scikit-Learn, Transformers, Pandas, Numpy, ONNX, OpenCV, spaCy, MatPlotlib, NLTK, Hadoop, XGBoost
Competencies: Vector databases, Machine Learning, CUDA, Artificial Intelligence, Data Science, Deep Learning, Data Analytics, Data Engineering, Prompt Engineering, Gen AI, Large Language Models, Statistical Modeling, Data Modeling, Statistics, Data Structures, Retrieval Augmented Generation, Recommendation System